



Constructing pie charts

- 1 36 people were asked: "what do you drink whilst at work". The results will be placed in a pie chart with a total of 12 pre-defined sectors. How many sectors would represent water? Tick 1 correct answer

	frequency	sectors
total	36	12
water	21	
milk		2
other		

☐ 21

☐ 63

☐ 12

☒ 7

☐ 2

- 2 36 people were asked: "what do you drink whilst at work". The results will be placed in a pie chart with a total of 12 pre-defined sectors. How many people voted for milk? Fill in the blank

	frequency	sectors
total	36	12
water	21	
milk		2
other		

6

- 3 A large set of information is collected and represented in a pie chart of 20 equally-sized sectors. Each one sector represents a frequency of 45. What is the total frequency of this dataset? Fill in the blank

900



- 4 The incomplete ratio table shows a dataset being converted into angles to represent each subgroup as sectors on a pie chart. Match the letters to the correct missing values. Write the correct letter in each box

	frequency	angle
forest	17	102
river	11	<i>d</i>
field	18	<i>e</i>
other	14	84
total	<i>a</i>	<i>b</i>

c

a	a
b	b
c	c
d	d
e	e

<i>b</i>	360
<i>e</i>	108
<i>a</i>	60
<i>d</i>	66
<i>c</i>	6

- 5 Starting with the first step, put these steps for constructing a sector of a pie chart in order. Use numbers to show the correct order

<i>5</i>	Plot an × at this point on the protractor.
<i>1</i>	Draw a straight line from the centre of the circle to its circumference.
<i>6</i>	Remove your protractor.
<i>3</i>	Line the 0° on the protractor with the most recent straight line drawn.
<i>4</i>	Starting from 0°, find the angle you want to draw on the protractor.
<i>7</i>	Draw a straight line from centre of the circle through the × you plot.
<i>2</i>	Line the bottom centre of the protractor with the centre of the circle.



- 6 The incomplete ratio table shows a dataset being converted into angles to represent each subgroup as sectors on a pie chart. Find the value of angle a , in degrees. Fill in the blank

	frequency	angle
white	6	
wholemeal	12	
granary	14	a
wraps	16	
total		360

105

